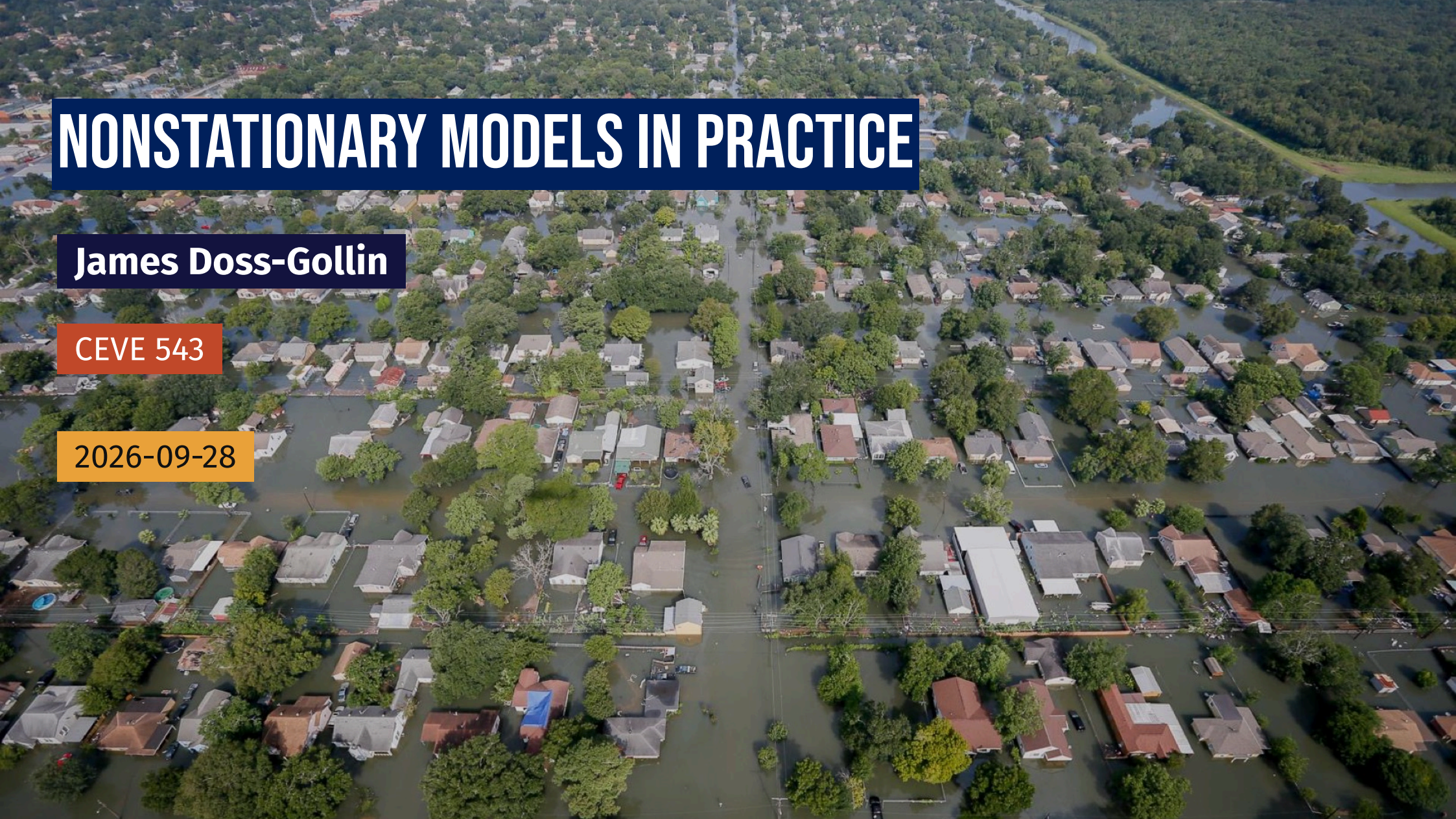
NONSTATIONARY MODELS IN PRACTICE

James Doss-Gollin

CEVE 543

2026-09-28



THE DATA

```julia
here = @__DIR__
rng = MersenneTwister(543)

hobby = CSV.read(joinpath(here, "hobby-annmax.csv"), DataFrame)
hobby = hobby[hobby.n_valid_days .>= 350, :]     # the complete years
co2 = CSV.read(joinpath(here, "logCO2.csv"), DataFrame)
hobby = innerjoin(hobby, co2; on=:year)           # adds a log_CO2 column
y = hobby.prcp_mm
```

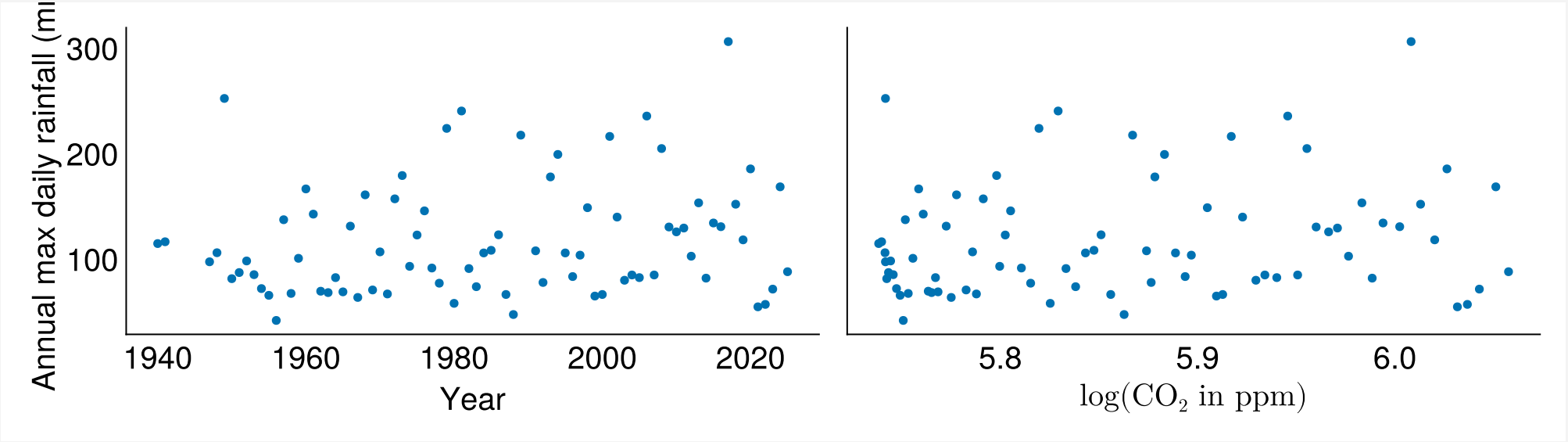
Houston Hobby annual maximum daily rainfall, 80 years, and the natural log of atmospheric CO2 in parts per million.

Choosing a covariate

Writing the model

Projecting

HOBBY ANNUAL MAXIMA AGAINST TWO COVARIATES



Think

Which covariate will fit better?

OVER THE RECORD, YEAR AND LOG CO2 MOVE TOGETHER

- Correlation 0.98: a fit on this record cannot tell them apart.
- They differ once you extrapolate, which is where the choice matters.

CHOOSE A COVARIATE BY MECHANISM

The covariate-based recipe (Schlef et al., 2023, p. 8), where EVD is an extreme value distribution:

- “Determine suitable covariates based on causative mechanisms”
- “Fit an EVD, with parameters dependent on the covariates”
- “Create future projections by forcing the EVDs with projections of the covariates”

“Time, as a covariate, does not represent a mechanistic process” (Schlef et al., 2023, p. 9).

THREE KINDS OF COVARIATE

Physical drivers used as covariates for extreme rainfall (from a paper in preparation in my group):

- **Thermodynamic:** temperature, such as global mean surface temperature.
- **Dynamical:** large-scale patterns that steer storms, such as El Niño (Niño 3.4).
- **External forcing:** CO2 concentration, used here and in Friday's lab.

WHICH COVARIATES PUBLISHED STUDIES TEST

Of the 16 covariate-based rainfall studies in the review (Schlef et al., 2023, Table 1):

- 12 test time, and 8 use nothing else.
- 6 test a climate index such as El Niño, and 4 test a temperature.
- None uses CO2.

KENDALL'S τ AS A COVARIATE SCREEN

Mann-Kendall's statistic is Kendall's τ between a record and the year. Put a covariate in place of the year and the same statistic screens the covariate.

i Note

Figure shown in class: Kendall's rank correlation between smoothed global mean surface temperature and 1-hour (left) and 1-day (right) annual maximum rainfall at Texas stations with at least 60 years of record; blue is positive and red negative, and most stations are positive (preliminary, from a paper in preparation in my group).

Choosing a covariate

Writing the model

Projecting

CENTERING AND SCALING THE COVARIATE

$$x_t = \frac{\log \mathrm{CO}_{2,t} - \overline{\log \mathrm{CO}_2}}{s_{\log \mathrm{CO}_2}}$$

```
1 x_center, x_scale = mean(hobby.log_CO2), std(hobby.log_CO2)
2 x = (hobby.log_CO2 .- x_center) ./ x_scale
```

- Uncentered, the intercept is the location at the year 0, or at 1 ppm of CO2: −128 mm and −59 mm.
- Centered, the intercept is the location in an average year, and the slope is per standard deviation, so priors are easy to set.
- Centering also makes the optimizer and the sampler faster.

A TREND IN THE LOCATION

$$y_t \sim \text{GEV}(\mu_0 + \mu_1 x_t, \sigma, \xi)$$

```
@model function gev_location_trend(y, x)
    μ0 ~ Normal(100, 50)        # mm, in an average year of the record
    μ1 ~ Normal(0, 25)          # mm per standard deviation of x
    logσ ~ Normal(3, 2)
    ξ ~ Normal(0, 0.25)
    σ := exp(logσ)
    for t in eachindex(y)
        y[t] ~ GEVKernel(μ0 + μ1 * x[t], σ, ξ)
    end
end
location_model = gev_location_trend(y, x)
```

This is the week 4 model with μ replaced by a regression on x_t.

MAXIMUM LIKELIHOOD FIT OF THE LOCATION TREND

```
start = InitFromParams((μ0=median(y), μ1=0.0, logσ=log(std(y)), ξ=0.1))
location_mle = maximum_likelihood(location_model; initial_params=start)
θ̂ = NamedTuple(location_mle.params)
@printf("μ0 = %.1f mm, μ1 = %.2f mm per sd, σ = %.1f mm, ξ = %.2f\n", θ̂.μ0, θ̂.μ1, θ̂.σ, θ̂.ξ)
```

```
μ0 = 90.8 mm, μ1 = 2.52 mm per sd, σ = 34.5 mm, ξ = 0.19
```

A TREND IN THE LOCATION AND THE SCALE

$$y_t \sim \text{GEV}\left(\mu_0 + \mu_1 x_t, \exp(\lambda_0 + \lambda_1 x_t), \xi\right)$$

```julia
@model function gev_location_scale_trend(y, x)
    μ0 ~ Normal(100, 50)
    μ1 ~ Normal(0, 25)
    λ0 ~ Normal(3, 2)           # log scale in an average year; exp keeps the scale positive
    λ1 ~ Normal(0, 0.5)         # change in log scale per standard deviation of x
    ξ ~ Normal(0, 0.25)
    for t in eachindex(y)
        y[t] ~ GEVKernel(μ0 + μ1 * x[t], exp(λ0 + λ1 * x[t]), ξ)
    end
end
start = InitFromParams((μ0=median(y), μ1=0.0, λ0=log(std(y)), λ1=0.0, ξ=0.1))
both_mle = maximum_likelihood(gev_location_scale_trend(y, x); initial_params=start)
```

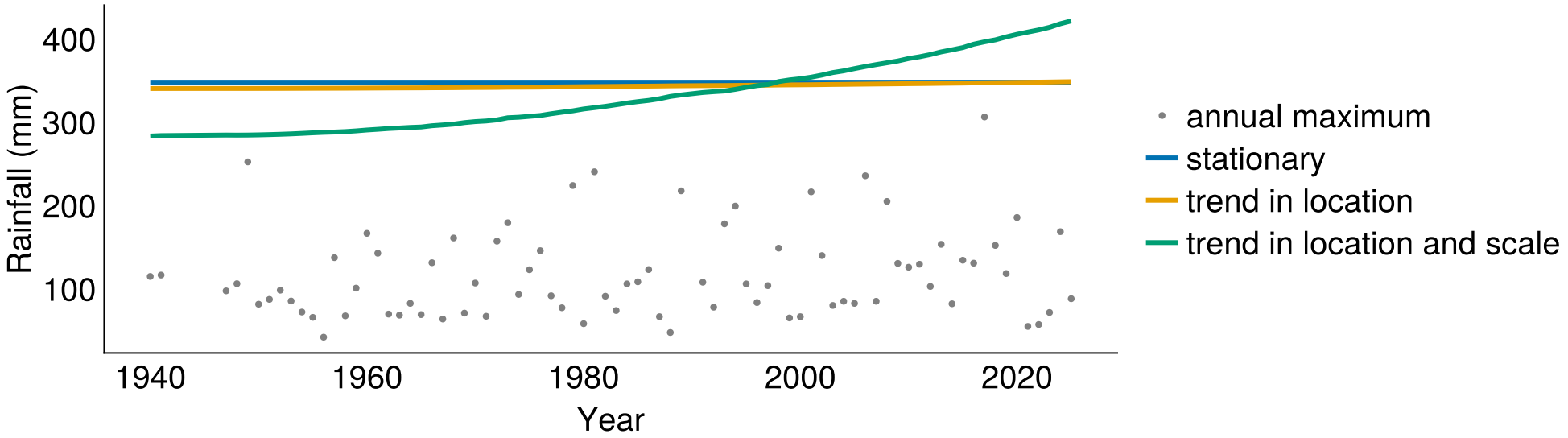
FITTING THE SAME MODELS WITH GEVFIT

```julia
# gevfit centers and scales covariates itself; gevfitbayes takes the same keywords
fit_location = gevfit(y; locationcov=[hobby.log_CO2])
fit_both = gevfit(y; locationcov=[hobby.log_CO2], logscalecov=[hobby.log_CO2])
fit_year = gevfit(y; locationcov=[hobby.year])
@printf("location: %.2f, %.2f\n", location_mle.lp, loglike(fit_location))
@printf("both:     %.2f, %.2f\n", both_mle.lp, loglike(fit_both))
@printf("year:     %.2f\n", loglike(fit_year))
```

```
location: -418.40, -418.40
both:     -417.42, -417.42
year:     -418.40
```

- Log likelihoods by hand and from `gevfit` match, and year gives the same fit as log CO2.

THE 100-YEAR LEVEL UNDER THREE MODELS



Adding the scale trend raises the log likelihood by 1.0 and moves the 2025 100-year level by 73 mm. Wednesday's reading makes the same comparison for Brays Bayou (Smith et al., 2026).

Choosing a covariate

Writing the model

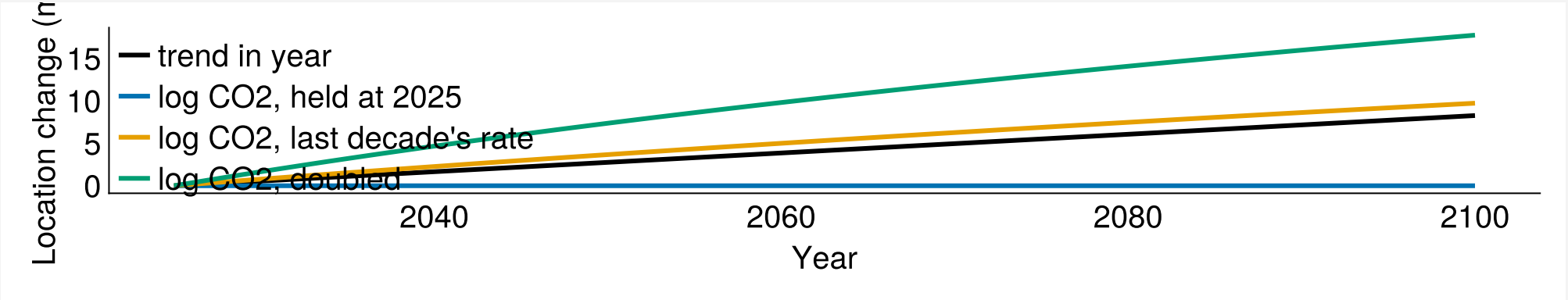
Projecting

THREE CO2 PATHS TO 2100

```
ppm_2025 = exp(last(co2.log_CO2))
recent_rate = (ppm_2025 - exp(co2.log_CO2[end - 10])) / 10     # ppm per year, 2015-2025
ppm_2100 = [
    "held at 2025" => ppm_2025, "last decade's rate" => ppm_2025 + 75recent_rate,
"doubled" => 2ppm_2025,
]
```

The three paths span a range and are not published emissions scenarios.

YEAR AND LOG CO2 EXTRAPOLATE DIFFERENTLY



A trend in year gives one projection. A trend in log CO2 gives a different projection for each CO2 path.

TWO SOURCES OF UNCERTAINTY IN A PROJECTED LEVEL

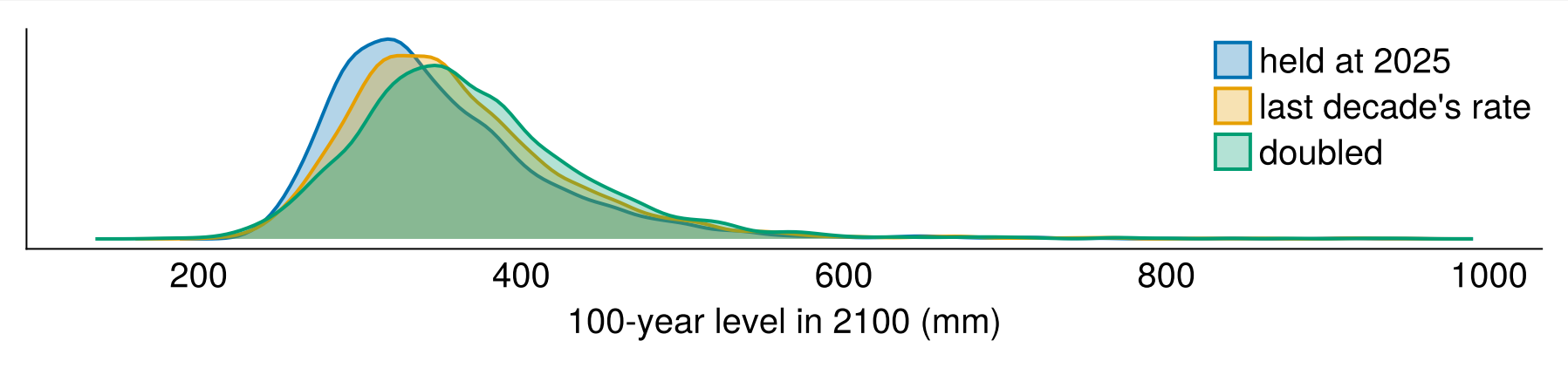
Let z be the 100-year level in 2100 and s the CO2 path. By the law of total variance,

$$\mathrm{Var}(z) = \underbrace{E[\mathrm{Var}(z \mid s)]}_{\text{parameters}} + \underbrace{\mathrm{Var}(E[z \mid s])}_{\text{CO2 path}}$$

```
chain = sample(rng, location_model, NUTS(0.99), MCMCThreads(), 1_000, 4; progress=false)
draws = DataFrame(chain)
level_100yr(μ, σ, ξ) = quantile(GeneralizedExtremeValue(μ, σ, ξ), 1 - 1 / 100)
x_2100(ppm) = (log(ppm) - x_center) / x_scale
levels_2100 = [label => [level_100yr(d.μ0 + d.μ1 * x_2100(ppm), d.σ, d.ξ) for d in eachrow(draws)] for (label, ppm) in ppm_2100]
```

Weighting the three paths equally is itself an assumption.

MOST OF THE UNCERTAINTY COMES FROM THE PARAMETERS




```julia
var_parameters = mean(var(z) for (_, z) in levels_2100)
var_path = var([mean(z) for (_, z) in levels_2100]; corrected=false)
@printf("parameters %.0f mm², CO2 path %.0f mm², path share %.1f%%\n",
    var_parameters, var_path, 100var_path / (var_parameters + var_path))
```

```
parameters 5521 mm², CO2 path 66 mm², path share 1.2%
```

EVERY TREND PARAMETER COSTS PRECISION IN THE TAIL

- Each coefficient is estimated from the same 80 years, and its uncertainty carries into every return level.
- Without a covariate that represents a mechanism, “the more parsimonious stationary model should be assumed” (Serinaldi and Kilsby 2015, quoted in Schlef et al. (2023), p. 10).
- On Wednesday, the Brays Bayou 100-year interval widens when the trend parameters are added (Smith et al., 2026, p. 16).